FoodLoop: An Explainable Urgency-Aware Matching and Collection Prioritisation Framework for Surplus Food Redistribution

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***Abstract*—Surplus food is often discarded because a suitable recipient cannot be identified and reached before the collection deadline. This paper presents FoodLoop, a web-based prototype and design for coordinating surplus-food donations, together with a rule-based method for prioritising listings for a recipient. The method first rejects listings that cannot be collected, then scores the rest on deadline urgency, category fit, quantity fit and, when a reliable travel time exists, collection feasibility. Criteria that cannot be assessed are excluded and the remaining weights are rescaled; the assessed coverage is also reported because scores with different coverage may not be directly comparable. Each result carries reasons computed from the same values as its score. The public demonstration offers listing, discovery, matching, reservation, pickup-tracking and analytics views and keeps its data in the browser; a cloud mode with Supabase authentication is described but is not evaluated here. Existing platforms already provide listing, collection confirmation and reporting, and earlier research applies learning to volunteer engagement, so the contribution claimed is limited to a specific, auditable combination of eligibility rules, explained urgency scoring and outcome records. No experiment has been conducted; the paper gives a test plan with hand-computed expected values.**

***Keywords*—surplus food redistribution, food rescue, rule-based matching, explainable prioritisation, multi-criteria scoring, pickup-outcome tracking, decision support.**

I. INTRODUCTION

Food is lost or wasted at every stage of the supply chain, and part of what is discarded is still safe to eat. The FAO's global study [1] and the UNEP Food Waste Index Report 2024 [2] describe the scale of the problem and its environmental and social costs. Redistributing surplus to people who need it is one response, but time limits it: cooked and perishable food has a short usable life, so a donation helps only if a suitable recipient collects it before it spoils.

Coordination often relies on phone calls and messaging groups. A donor holding a tray of cooked meals and a two-hour window cannot easily see which nearby organisation accepts that kind of food, has room for it and can reach it in time. Recipients, for their part, see offers without any signal about which are most pressing.

FoodLoop targets this step. It is a decision-support design and prototype, not a transport service or a food-safety authority. It lists surplus, rejects listings a recipient cannot collect, ranks the remainder by urgency and fit, explains the ranking, and records reservations and pickup outcomes.

Section II states the research questions, Section III reviews related work, and Section IV states the gap and contributions. Sections V to VIII present the architecture, method, implementation status and data model. Section IX gives the evaluation plan, and Sections X to XIII cover discussion, limitations, future work and the conclusion.

II. PROBLEM STATEMENT AND RESEARCH QUESTIONS

Given active donation listings, each with a food category, quantity and unit, and a collection deadline, and recipient organisations, each with a stated category preference, a maximum collection quantity and a time from which they can collect, the system must decide which listings a recipient sees and in what order. Three difficulties shape the design.

1. **Time pressure.** A listing gains urgency as its deadline approaches, but a listing that cannot be reached before the deadline has no value, however urgent it appears.
2. **Heterogeneous needs.** A kitchen that serves only vegetarian meals, a shelter that can store frozen goods and a community fridge with limited space have different needs, so showing every listing to everyone adds noise.
3. **Accountability.** If a ranking cannot be explained, donors may distrust it and recipients cannot question it. Without outcome records, nobody can tell whether the process works.

The paper addresses three research questions:

- **RQ1.** Can a small rule-based method rank listings for a recipient so that every score and explanation can be reproduced by hand from the listing and profile data?
- **RQ2.** How should missing or incomparable data, such as an unknown travel time, a unit mismatch or an absent preference, be treated so that an absent input is not silently scored as favourable or unfavourable?
- **RQ3.** Which recorded outcomes are needed to evaluate such a method later?

III. RELATED WORK

The sources below were found through targeted searches and checked against publisher, library or institutional pages. The review is not systematic and no count of studies is claimed.

A. Scale of food waste. Gustavsson et al. [1] estimated global food losses and waste for the FAO and discussed their causes along the supply chain. UNEP's 2024 report [2] provides an updated global estimate and guidance on measuring and reducing food waste. Both motivate redistribution, but neither addresses the coordination of a single donation.

B. Digital food-waste platforms. Oroski and da Silva [3] reviewed platforms that reduce food waste and concluded that digital platforms are a promising option but that technology alone does not recover food. They identify barriers such as delivering benefits to heterogeneous actors and building trust, and a shortage of empirical studies of real impact. Operating platforms already cover core functions: the FAO AgriTech Observatory describes FoodCloud as a platform where donors upload surplus offers, charities accept and confirm collections, and donation data support operational and impact reporting [4]. Listing, collection confirmation and reporting are therefore established capabilities.

C. Learning-based coordination in food rescue. Shi et al. [5] proposed a recommender system for crowdsourced food rescue that selects which volunteers to notify about each rescue, and Shi et al. [6] introduced a bandit data-driven optimisation method for volunteer recommendation and showed that it outperformed an existing baseline. In both papers the platform already matches donations to recipient organisations, and the learning problem is the

engagement of the volunteers who transport the food. FoodLoop addresses an earlier step, which listings a recipient should see and in what order, and uses fixed rules. That suits a setting without historical data but offers no adaptation.

D. Allocation and matching theory. Gale and Shapley [7] introduced stable matching, and Kuhn [8] described a method for the assignment problem; both assume well-defined scores or preference orderings. Sengul Orgut et al. [9] modelled the equitable and effective distribution of donated food by a food bank as network-flow problems in which equity and effectiveness can conflict. The finding matters here because any ranking rule has distributional consequences.

E. Multi-criteria scoring and participation. The Analytic Hierarchy Process [10] and simple additive weighting express a decision as weighted criteria whose contributions can be inspected, at the price of judgement-based weights. Lee et al. [11] describe WeBuildAI, in which stakeholders build models of their own preferences that then determine algorithmic policy, and apply it to a food-donation matching setting. This suggests that FoodLoop's weights should eventually be set with stakeholder input.

F. Recommender systems. Latent-factor collaborative filtering [12] needs interaction history and suffers from cold start for new users and items, which fits poorly with short-lived, one-off listings.

IV. RESEARCH GAP AND CONTRIBUTIONS

The literature reviewed above establishes four points. Digital platforms for surplus food exist and have been reviewed [3]. Operating platforms already provide listing, collection confirmation and reporting [4]. Learning-based work in food rescue concentrates on volunteer engagement and depends on data and trained models [5], [6]. Allocation studies optimise a stated objective at food-bank level [9]. Among the sources reviewed, none describes a lightweight, rule-based design in which explicit eligibility rules precede an urgency score whose reasons can be reproduced by hand and whose results are tied to recorded pickup outcomes. Because the search was targeted, this is a limit of the review and not evidence that no such system exists.

The paper therefore proposes four design contributions. None is claimed as proven or as superior to existing systems.

1. A specification of eligibility rules and criterion functions, with defined behaviour for expired listings, invalid quantities, unit mismatches and missing preferences (Section VI).
2. A missing-data rule that drops unassessable criteria and rescales the remaining weights, and reports how much of the total weight was assessed.
3. Reasons generated from the same criterion values and weights that produce the score.
4. A reservation and pickup-outcome record that separates listed from collected quantity, with a hand-checkable test plan (Section IX).

V. SYSTEM ARCHITECTURE

The public demonstration page [13] presents FoodLoop as a single-page workspace. Its sections are Overview, Discover food, Saved food, Donate food, Smart matching, My reservations, Pickup tracking, Impact analytics, Admin dashboard and About, and a role selector offers Recipient, Donor and Administrator views. The page was read but not operated, and the architecture in Fig. 1 summarises what it states.

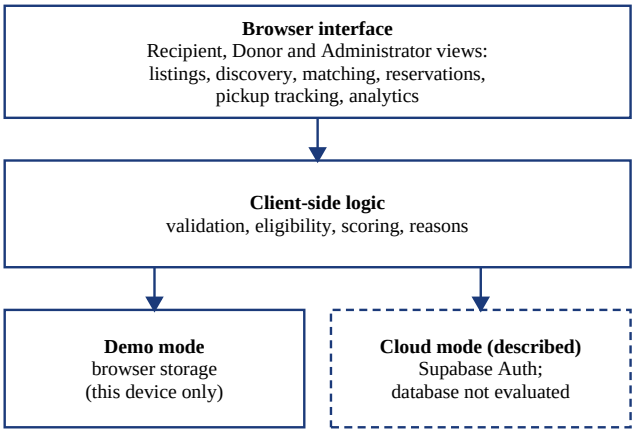


Fig. 1. Architecture as described on the demonstration page [13]. Dashed outline: described, not evaluated.

Listing and discovery. The donation form asks for a food name, a category (produce, bakery, prepared meals, dairy, pantry or other), a quantity with a unit (kg, items, portions, boxes or litres), a pickup location, an availability start, a collect-by deadline, storage information and contact details. Discovery offers a category filter and sorting by soonest deadline, most recent listing or largest quantity.

Matching. The Smart matching section accepts a sample or custom recipient profile with a preferred category, a maximum collection quantity with unit and a time from which the recipient can collect, and returns a prioritised list described as a rule-based engine. The page states that distance and travel time are not scored because no verified data are available.

Tracking and analytics. The workspace lists reservations with status filters, shows pickup tracking (in progress and completed), summarises impact by period with a downloadable report, and gives administrators a view of all donations. The page labels sample records as such.

Data. In demo mode the page states that changes stay in the browser. Browser storage is specific to one browser profile on one device, is not shared between users and is not a secure store, so it cannot support a multi-user deployment. The sign-in dialog states that cloud mode uses Supabase Auth [14]. This paper does not evaluate a database behind that mode.

VI. METHODOLOGY AND MATCHING ALGORITHM

This section specifies the proposed method. The four weights are those published on the demonstration page [13]. The eligibility rules, criterion functions, missing-data rule and tie-breaking below are the specification proposed in this paper, and Section VII states what has and has not been checked against the implementation.

A. Inputs and notation

For a listing l let Q_l be its quantity in unit u_l, d_l its collect-by time, a_l its availability start and c_l its category. A recipient r has an optional preferred category c_r, an optional maximum collection quantity Q_r in unit u_r, and an optional collection start t_r. Let t_{now} be the current time, T_{trav} an optional reliable travel-time estimate, and H a fixed urgency horizon. The usable start is $t_s = \max(t_{now}, a_l, t_r)$ and the remaining window is $T_{rem} = d_l - t_s$.

B. Validation and hard eligibility

A listing is rejected at entry if its deadline is missing, its quantity is not a positive number or its unit is missing. For a valid listing, a pair (l, r) is eligible only if all of the following hold, and ineligible pairs are never scored:

- the listing is still open (not reserved, cancelled or collected);

- $T_{rem} > 0$, so an expired listing or a zero remaining window is excluded;
- if a reliable T_{trav} exists, $T_{trav} < T_{rem}$, so a trip that uses the whole window is excluded.

A recipient maximum Qr must be positive. If Qr <= 0, the recipient cannot accept a positive quantity, so the pair is ineligible and not scored. If Qr is missing or its unit differs from the listing unit, quantity fit is unassessed and its weight is excluded; remaining weights are rescaled. A smaller donation is not penalised, while excess lowers the score (Eq. 3).

C. Criteria

Each assessable criterion yields a value in [0, 1]. Urgency is always assessable once a pair is eligible.

$$f_{urg} = 1 - min(1, T_{rem} / H) \quad (1)$$

Because $T_{rem} > 0$ and $H > 0$, f_{urg} lies in [0, 1) and is largest when little time remains.

$$f_{cat} = 1 \text{ if } c_l = c_r, \text{ otherwise } 0 \quad (2)$$

Category fit is assessed only if the recipient states a preference. Choosing any category leaves it unassessed; it is not scored as a perfect match.

$$f_{qty} = min(1, Q_r / Q_l) \quad (3)$$

Quantity fit is assessed only if both quantities are present and $u_l = u_r$. It is 1 when the whole listing fits within the maximum and falls as the listing exceeds what the recipient can take.

$$f_{feas} = (T_{rem} - T_{trav}) / T_{rem} \quad (4)$$

Feasibility is assessed only if a reliable T_{trav} is available. Since $T_{trav} < T_{rem}$ for eligible pairs, f_{feas} lies in (0, 1]. A negative or unreliable T_{trav} is treated as missing.

D. Score and missing-data rule

Let A be the set of assessable criteria for the pair, with base weights $w_{urg} = 0.40$, $w_{cat} = 0.25$, $w_{qty} = 0.20$ and $w_{feas} = 0.15$, which sum to one. The base weights are judgement-based, were not learned and have not been validated. The weights of assessable criteria are rescaled and combined:

$$S(l, r) = 100 \times \Sigma_{k \text{ in } A} w'_k f_k, \; w'_k = w_k / \Sigma_{j \text{ in } A} w_j \quad (5)$$

Since each f_k lies in [0, 1] and the rescaled weights sum to one, S lies in [0, 100]. The coverage $C = 100 \times \Sigma_{k \text{ in } A} w_k$ is reported with the score, because scores computed over different sets A are not strictly comparable. A is never empty, since it always contains urgency; its minimum coverage is 40%.

E. Reasons and ordering

For every criterion in A, the displayed reason states the input values and the points contributed, $100 \times w'_k f_k$, and the points sum to S. Criteria outside A are listed as not assessed, with the cause. Because the reasons use the same f_k and w'_k as the score, they cannot diverge from it. Pairs are ordered by S, then by earlier d_l, then by earlier listing time, then by listing identifier; this tie-breaking rule is proposed. Algorithm 1 summarises the procedure.

Algorithm 1. Ranking listings for one recipient (proposed)

```
for each valid, open listing l:
  ts = max(now, a_l, t_r);  Trem = d_l - ts
  if Trem <= 0: skip
  if Ttrav is reliable and Ttrav >= Trem: skip
  f = {urg: 1 - min(1, Trem/H)}
  if Qr <= 0: skip; if Qr present and units match: fqty=min(1,Qr/Ql)
  W = sum of base weights over keys of f
  S = 100 * sum(w[k]/W * f[k] for k in f)
  reasons = points 100*w[k]/W*f[k]; unassessed keys
sort by S desc, d_l asc, listed_at asc, id asc
```

F. Edge cases and worked example

Table I lists how each edge case is handled. The example that follows is hypothetical and illustrates the arithmetic; it is not a result.

TABLE I. EDGE-CASE HANDLING

Case	Treatment
Expired listing; $T_{rem} \le 0$	Ineligible; not scored
Missing deadline	Listing rejected at entry
Quantity zero, negative or non-numeric	Listing rejected at entry
Max <= 0; missing; unit mismatch	<= 0: ineligible; otherwise unassessed
No category preference	Category criterion not assessed (not full credit)
Travel time missing, negative or unreliable	Feasibility not assessed
$T_{trav} \ge T_{rem}$	Ineligible; not scored
Only urgency assessable	Scored on urgency alone; coverage 40%

Suppose 2 h remain (T_{rem} = 120 min), H = 12 h, the category matches the recipient's preference, the listing is 60 kg against a 50 kg maximum, and travel takes 20 min. Then $f_{urg} = 1 - 2/12 = 0.833$, $f_{cat} = 1$, $f_{qty} = 50/60 = 0.833$ and $f_{feas} = (120 - 20)/120 = 0.833$. All four criteria are assessed, so $S = 100 \times (0.40 \times 0.833 + 0.25 \times 1 + 0.20 \times 0.833 + 0.15 \times 0.833) = 87.5$, from 33.3, 25.0, 16.7 and 12.5 points. Without travel data, A has weight sum 0.85 and $S = 100 \times (0.333 + 0.25 + 0.167)/0.85 = 88.2$ at 85% coverage.

VII. TECHNOLOGY STACK AND IMPLEMENTATION STATUS

Tables II and III record what the public page [13] states. The page was read on 16 October 2026 but not operated, and its source code was not inspected, so no entry is confirmed by execution. The base weights in Section VI match those the page publishes. Whether the eligibility rules, criterion functions, missing-data rule and ordering in Section VI match the deployed code has not been checked, and until it is, they stand as the specification proposed here.

TABLE II. TECHNOLOGY STATUS

Technology	Role	Basis	Status
HTML, CSS, JavaScript	Client-side workspace	Page retrieved; code not inspected	Unverified
Netlify hosting	Public demonstration	Page served from a netlify.app address	Observed once
Browser storage	Demo-mode persistence on one device	Page states that changes stay in the browser	Stated, not tested
Supabase Auth	Cloud-mode sign-in [14]	Sign-in dialog text	Stated, connection not tested
Database and row-level security	Shared storage and access control [14], [15]	None	Not evaluated
Maps and travel time	Feasibility input	Page says geography is not scored	Not implemented
Machine learning, notifications, external APIs	None	Page disclaims machine learning; none seen	Not found

TABLE III. FEATURE STATUS

Feature	Basis on page	Status
Listing form with category, quantity and unit, location, availability, deadline, storage and contact	Form fields; required fields marked	Seen in UI, not tested
Discovery filters and sorting	Category filter; three sort options	Seen in UI, not tested
Recipient profile; ranked list with reasons	Smart matching section; weights shown	Seen in UI; logic not inspected
Eligibility rules; duplicate-reservation rule	Not shown	Not verified
Reservations; pickup tracking	Reservation table with status filters; progress views	Seen in UI, not tested
Analytics and report download	Period filter; outcome panels	Seen in UI, not tested
Roles and permissions	Role selector; Supabase sign-in text	Partly described; not verified
Mobile layout; accessibility	Viewport tag only	Not verified

VIII. DATA MODEL

Table IV lists the records that the method and the outcome analysis need. It is a specification of required data, not a description of the deployed schema. In demo mode such records would be held as JSON in browser storage. A shared deployment would need a database with a uniqueness rule that allows one active reservation per listing and row-level access policies [15].

TABLE IV. RECORDS REQUIRED BY THE METHOD

Record	Fields used	Purpose
Listing	id, category, quantity, unit, available_from, collect_by, location, status, listed_at	Eligibility and scoring inputs
Recipient profile	id, preferred_category, max_quantity, unit, collect_from, location	Scoring inputs
Reservation	id, listing_id, recipient_id, reserved_at, status	One active reservation per listing
Pickup outcome	id, reservation_id, outcome, collected_quantity, recorded_at	Collected, missed or cancelled; analytics

IX. TESTING AND EVALUATION PLAN

No test has been run for this paper, so no result is reported. Table V gives test cases with expected outcomes derived by hand from Section VI; each can be executed against the scoring function and the workflow. Any dataset used for scenario tests must be labelled synthetic, and synthetic outcomes must not be presented as field results.

TABLE V. TEST CASES AND EXPECTED OUTCOMES

ID	Input	Expected outcome
T1	$T_{rem} = 0$ or negative	Pair not shown
T2	$T_{trav} = T_{rem}$	Pair not shown
T3	Worked example, Section VI-F	S = 87.5; points 33.3, 25.0, 16.7, 12.5; coverage 100%
T4	T3 without travel time	S = 88.2; coverage 85%; feasibility listed as not assessed
T5	T3 without travel time or category preference	S = 83.3; coverage 60%
T6	Recipient max <= 0; or units mismatch	<= 0: ineligible; mismatch: quantity unassessed; weights rescaled
T7	Quantity 0 or −5; missing deadline	Listing rejected at entry
T8	$Q_l \le Q_r$	$f_{qty} = 1$; no shortfall penalty
T9	Same listing with smaller T_{rem}	f_{urg} increases
T10	Any scored pair	Displayed points sum to S within rounding
T11	Second reservation of a reserved listing	Rejected
T12	Collected, then set to reserved	Invalid transition rejected
T13	Reload in the same browser; open in another	Data persist in the first; absent in the second
T14	Synthetic outcome set	Analytics counts and collected quantity equal the records
T15	Weights varied by 0.1, then rescaled	Record how often the top-ranked listing changes

For scenario comparison, the ranking can be compared with first-come and soonest-deadline-first orderings on synthetic data, using the number of listings that would pass their deadline unclaimed and the rank agreement between methods. Permission tests apply only if cloud mode is evaluated. A field pilot would need partner consent and ethical approval where required, and would record collection rate and time to collection from the outcome records.

X. DISCUSSION

The central design choice is to treat eligibility as a gate and urgency as a ranking signal. Urgency alone would place an almost-expired but unreachable listing first. Applying the gate first guarantees that this cannot happen, at the cost of excluding pairs that a lenient rule might still show.

The second choice is to treat missing data as unassessed, not as zero or full credit. Travel time is the input most likely to be missing in a prototype, and the demonstration page already leaves geography unscored and says so. Rescaling the weights keeps the score in range, and reporting coverage tells the reader how much of the weight was actually assessed. The consequence is that scores with different coverage are not strictly comparable.

Urgency and feasibility pull in opposite directions: little remaining time raises urgency but leaves little slack for travel. The eligibility gate removes the extreme case, and the sensitivity test T15 is intended to show whether the remaining interaction matters.

Outcome records matter because, as reviews of platforms note [3], evidence of real impact is scarce. Recording collected quantity separately from listed quantity allows later analysis to rest on what was collected. Global assignment [7], [8] could allocate more efficiently when preferences and capacities are known, and learning-based methods [5], [6] could adapt to data; both are natural comparison points once data exist.

XI. LIMITATIONS AND ETHICAL CONSIDERATIONS

Limitations. No evaluation has been done. The base weights are judgement-based and unvalidated, and the correspondence between the specification and the deployed code is unchecked. The demonstration keeps data in one browser, geography is not scored, and the literature search was targeted, not systematic.

Food safety. FoodLoop does not inspect, certify or guarantee food, and the demonstration page says so [13]. In India, the Food Safety and Standards (Recovery and Distribution of Surplus Food) Regulations, 2019 were notified by FSSAI [16]. They specify the responsibilities of food donors and surplus-food distribution organisations, prohibit a food business operator from passing on unsafe food, and require surplus food to be handed over early enough to be consumed within its shelf life. They also set labelling and record-keeping requirements [16]. FoodLoop does not check compliance, and its listing fields are not claimed to meet those requirements, so these responsibilities rest with donors and recipient organisations.

Fairness. Urgency-based ranking may favour recipients who check the platform often or who have better connectivity or transport. The distribution of collected food across recipients should be monitored, in line with the equity concerns in [9], and a recipient-side delay or notification mechanism may be needed.

Privacy and security. Donor and recipient contact details are personal data. The page advises donors to share only contact details they are content to make available to recipients [13]. Browser storage is neither shared nor secure, and this paper does not evaluate any access control. A shared deployment should collect only necessary data, enforce role-based access, obtain consent and follow applicable data-protection law. These are recommendations, not implemented safeguards.

Pilot ethics. A pilot needs institutional approval where required and informed participation by partner organisations.

XII. FUTURE WORK

- Check the specification in Section VI against the implementation, then run the tests in Table V.
- Evaluate cloud mode with shared storage, authentication and row-level access control.
- Add verified travel-time estimates so that feasibility can be assessed.
- Set weights with donor and recipient input, following participatory approaches [11].
- Compare with first-come, soonest-deadline-first, assignment-based [7], [8] and learning-based [5], [6] baselines on synthetic and, with consent, partner data.
- Monitor fairness of collected food across recipients.

XIII. CONCLUSION

FoodLoop proposes a transparent, rule-based method for coordinating surplus food: eligibility rules, a score that drops unassessable criteria and reports its coverage, reasons computed from the score, and reservation and pickup records. Earlier work shows that platforms, learning-based volunteer engagement and food-bank allocation models already exist, so the contribution is limited to this specific combination. Its usefulness has not been tested, and the evaluation plan in Section IX defines the evidence needed.

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